Condition-Preserving Metabolomics Batch Correction

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Purpose and intended demonstration

Estimate technical batch offsets while retaining a fictional biological condition effect. This is a computational biology learning project with synthetic or fictional inputs, not a clinical or experimentally validated system.

How the implementation works

Fit each feature against intercept, condition and batch indicators on training samples. Subtract only fitted batch offsets. Preserve the condition component rather than centering entire groups indiscriminately.

Evaluation and evidence

Use held-out samples from known batches and compare estimated technical-effect removal with naive centering. Hidden clean profiles are available only because the data is synthetic.

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Setup and commands

Requires Python 3.10 or newer and NumPy. Create a virtual environment: python -m venv .venv. On Windows use .venv\Scripts\python.exe if activation is unavailable. Install with python -m pip install -r requirements.txt. Run python app.py, then open results/report.html in your browser. Run python -m unittest discover -v to execute tests. The default seed is 42; change it with --seed. Replace sample_input.json using --input path-to-file.json, preserving its schema. Algorithm-only projects generate synthetic arrays from the seed rather than consuming external files.

Architecture and file responsibilities

CLI arguments -> JSON input or seeded generator -> domain pipeline -> validation or evaluation -> JSON and HTML artifacts. Imported functions can be used in a notebook or a separate application. Model checkpoints preserve learned arrays where applicable; see app.py for parameter order. This package does not include a production web service. Each project is independent and includes its own helper code so the ZIP can be extracted and run alone. app.py holds the experiment and domain functions: design, fit, correct, run. kit.py contains reporting, input loading and only the retrieval or generation helpers needed by this project. sample_input.json contains editable inputs. test_app.py checks domain behaviors; test_reporting.py checks metrics and HTML output boundaries. results/metrics.json records outputs and results/report.html provides a browser-readable report. Numeric projects also save arrays or learned parameters to results/artifacts.npz; inspect its keys with numpy.load. MODEL_CARD.md explains intended use and limits. .github/workflows/tests.yml runs tests and the fixture experiment. No API key is needed.

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Limitations and next improvements

Condition labels are used for fitting training batch effects; test condition labels are not used by correction. New unseen batches and confounded batch-condition designs need separate treatment. Check design identifiability, evaluate biological signal preservation, include pooled controls and compare established methods on a licensed metabolomics dataset.

Interview walkthrough

Present the problem and demonstrate one normal example, then one failure or boundary case. Explain the baseline or fixture mode, the split or validation design, and one limitation before discussing an extension. Be ready to trace data through the functions listed above. Explain why the chosen metric matches the task and why the available results cannot establish production performance. Add your own documented experiment and be precise about what you implemented or changed. These are AI-assisted educational projects; filenames and synthetic scores are not evidence of client work.

Research context and reading

These topics were selected from research directions reviewed on 22 October 2026. Recent activity suggests useful skills to develop; it is not a forecast of hiring, commercial success or scientific validity. The implementation is a small educational baseline and does not reproduce or claim the performance of any cited foundation model. Reading: Robust metabolomics normalization across experimental designs (2026): https://pubs.acs.org/doi/10.1021/acs.analchem.5c06841.

Verified build results

project

"Condition-Preserving Metabolomics Batch Correction"

raw to clean rmse

1.0463829990425848

corrected to clean rmse

0.018722435714400975

naive centering to clean rmse

0.05581407398919997

training samples

600

test samples

200

test batch labels known

true

test condition labels used for correction

false

scope

"Known-batch synthetic correction; no disease interpretation"

See results/metrics.json for all observations and TEST_RESULTS.txt for the executed tests. All datasets and replay responses shipped here are synthetic. Results can vary across seeds and numerical library versions.